YASH RAJ

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RTL/Digital Design Engineer | ASIC Front End (RTL to GDSII) | Cadence Xcelium · Genus · Tempus · Innovus | Verilog/SystemVerilog | CSP Scholar (Top 1%)

Education

IIIT-D (Indraprastha Institute of Information Technology, Delhi) Delhi
Bachelor of Technology, Electronics & Communication Engineering *2021 - 2025*

Relevant Coursework

- Digital Circuits
- Circuit Theory
- CMOS
- Digital VLSI Design
- FPGA Design
- Semiconductor Devices
- VLSI Design Flow
- Computer Architecture

Work Experience

OrVis Semi (CMOS Imaging Semiconductor Startup) | FPGA/RTL Engineer Oct 2025 - Present
AR0130 Camera Pipeline – Real-Time Bayer-to-UVC Video Streaming

- Architected a real-time **Verilog** video pipeline on Spartan-7 streaming live 1284×968 12-bit Bayer from an AR0130 **CMOS sensor** to a PC as **UVC** via FX3 **USB 3.0** – 5-stage bilinear **demosaicing** (BRAM line buffers, 3×3 window, stall-free), Q8 **fixed-point** white balance, and BT.601 RGB-to-**YUV422** conversion at ∼35 fps, conforming to externally-owned firmware geometry.
- Bridged asynchronous 50 MHz pixel and 100 MHz **USB** clock domains via an 18-bit FWFT async **FIFO** (**CDC**) with in-band frame/line sync; closed **5 hardware bugs** via Xilinx **ILA** and counter readback, including a USB de-enumeration fault resolved through output clamping.

SD Card Controller using SPI on FPGA (Spartan-7)

- Implemented a **multi-module Verilog** SPI-mode SDXC controller on Xilinx Spartan-7 full initialization sequence (CMD0→CMD8→ACMD41 HCS=1→CMD58 OCR/CCS), CMD17/CMD18 block reads with CMD12 stop, and CMD24 single-block write; hardware-validated via write→read→byte-level data integrity check at **12.5 MHz**.
- Designed shared-SPI architecture with runtime-configurable master (200 kHz init / 12.5 MHz data), **FSM**-arbitrated ownership across 4 modules, 2-stage MISO **CDC synchronizer**, and 512-byte shared **BRAM** buffer; debugged using Xilinx **ILA**, UART hex dump, and LED-mapped status reporting.

Projects

Network on Chip: RTL to GDS Flow Jan 2025 - Apr 2025

- Architected a 2×2 NoC in Verilog (4 modules: mesh, master, router, ProcessingUnit); developed 3 directed **Verilog testbenches** in Cadence **Xcelium** achieving **100% block coverage** and **80% code coverage**.
- Synthesized RTL using Cadence **Genus** (90 nm library) achieving timing closure at 2.55 ns (**WNS +814 ps**, 1649 cells); verified RTL-to-netlist equivalence using Cadence **Conformal** across all synthesis scenarios.
- Completed physical design in Cadence **Innovus** - placement, **CTS**, 9-layer routing at 0.5/0.8 utilization; post-route timing closure (setup +110 ps); **DFT** scan insertion and **STA** via Cadence **Tempus**, culminating in clean **GDSII**.

3×3 NoC Router - Cycle-Accurate Simulator Jul 2024 - Dec 2024

- Architected a cycle-accurate **Python** simulator for a 3×3 NoC mesh (9 routers) modeling input buffers, Switch Allocator (SA), and Crossbar (XBAR); implemented **XY/YX** routing with HF/BF/TF flit-level packet modeling via command-line configurability.
- Engineered dual simulation modes - **PVA** (nominal delays) and **PVS** (Gaussian variation, $\sigma = 10\%$, $\pm 3\sigma$ spread) - generating cycle-accurate log files, per-packet latency comparison reports, and flits-per-connection bar graphs.

Technical Skills

HDLs & Languages: Verilog, SystemVerilog, C++, Python, Tcl Scripting

Digital Design: RTL Design, FSM Design, Pipelining, Fixed-Point Arithmetic, Clock Domain Crossing (CDC), Asynchronous FIFO Design, BRAM, Low Power Design, AMBA (AXI, APB), SPI, UART, I2C

Verification: Directed Testbench, Functional Simulation, Hardware Validation, Formal Equivalence Checking

ASIC Flow & Tools: RTL-to-GDSII, Synthesis (Genus), Static Timing Analysis (Tempus), Place & Route (Innovus), Clock Tree Synthesis (CTS), Cadence Xcelium, Conformal

FPGA & Debug: Xilinx Vivado, Xilinx Spartan-7, XDC Constraints, Integrated Logic Analyzer (ILA), Keysight DSO

Other Tools: Git, GitHub, VS Code

Scholarships, Research & Leadership

Cadence Scholarship Program by Cadence Design Systems: CSP Scholar (awarded to **top 1%** ECE students.)
Undergraduate Researcher, Cognitive Sciences Lab, IIIT Delhi: Developed Python and MATLAB-based signal processing pipelines for real-time EEG data acquisition, saliency modeling, and eye-tracking analysis.